CALCULATOR PRO

A multi- purpose Desktop calculator built l with python and Tkinter



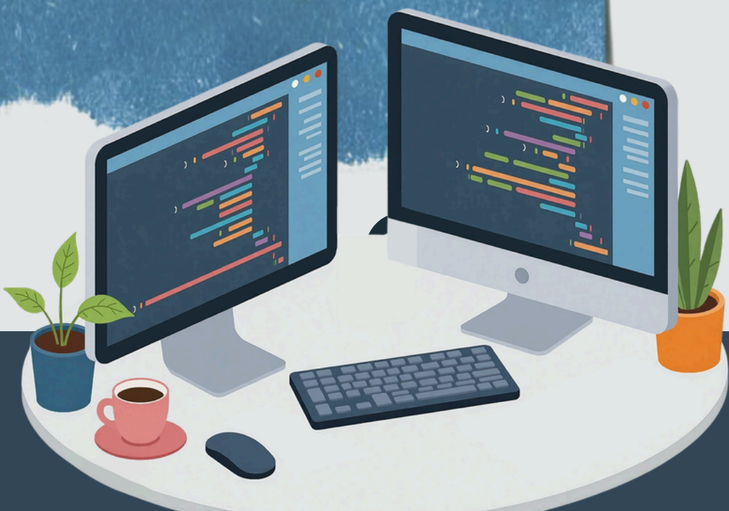
Presented by Hemlata

Presented to NEXTHIKES IT SOLUTION

Introduction

A feature-rich desktop calculator developed using Python and Tkinter, combining mathematical operations, a graphical user interface, memory management, calculation history, keyboard support, and dark/light themes with safe expression handling.



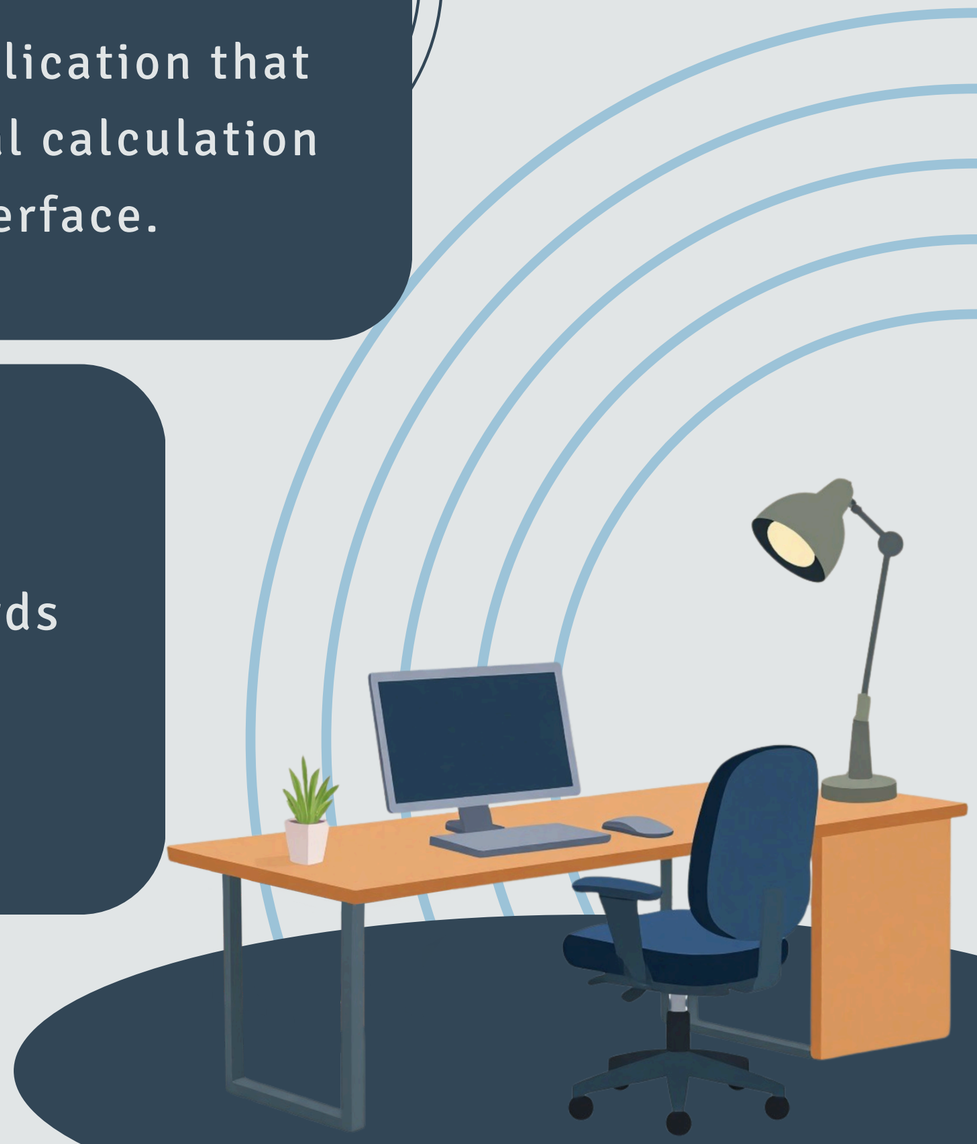


Project Overview

calculator Pro is desktop application that perform common mathematical calculation through a graphical interface.

Core idea

- Enter number using on-screen buttons or keybords
- View the result in the display area
- Switch between dark and light themes



Key Features

Basic Operations

Addition,subtraction, multiplication and division

Extra calculations

Square root, square, power and percentage

Calculation history

Store completed calculations and display them in a separate window

Themes

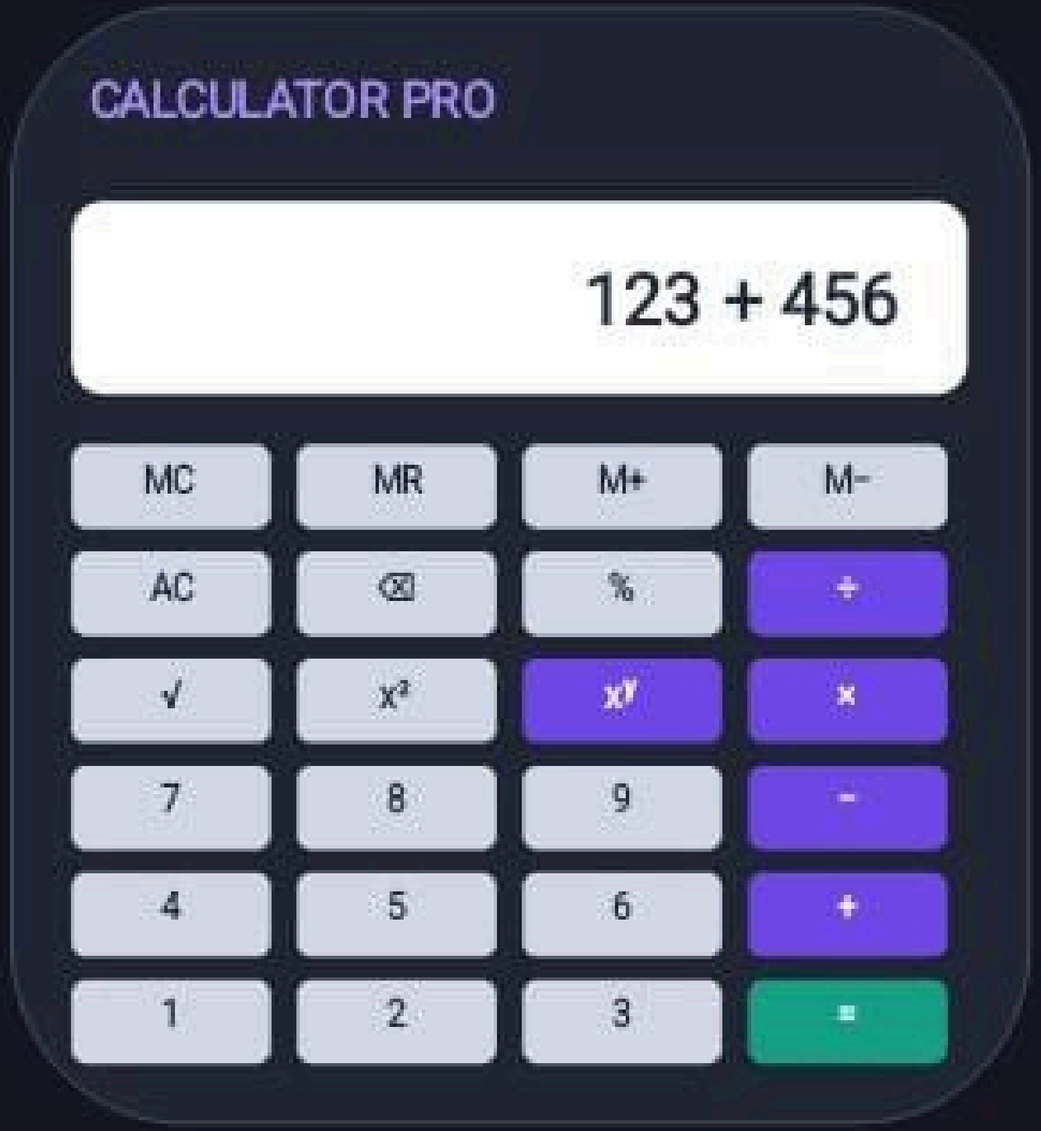
Switches between dark mode and light mode

Editing controls

AC clears the display; backspace removes the last character



User Interface



interface components

- Header: application title and theme toggle
- Display: shows the current expression and result
- Button grid: number, keys, operators, and special functions
- History button: open saved calculation records
- Responsive layout: window has a minimum size

Technologies Used

Python

Main programming language used to write the application

Tkinter

creates the desktop window, display, buttons, labels and history window

ast-operator

Reads arithmetic expressions as syntax and applies approved operations

math

Provides mathematical function such as square root and finite-number checks.

Python functions

Organise tasks such as calculation, clearing, memory, theme, and button actions

KEY HIGHLIGHTS

- Basic Arithmetic Operations (+, −, ×, ÷)
- Square Root, Square & Power
- Percentage Calculation
- Memory Functions (MC, MR, M+, M−)
- Dark & Light Mode
- Calculation History
- Keyboard Support
- Error Handling
- User-Friendly Graphical Interface (GUI)



conclusion

Calculator Pro demonstrates my practical understanding of Python and Tkinter. This project helped me develop programming skills and create a simple, interactive, and user-friendly calculator.



Thank You

